

Session 9 Quiz - AI Hallucinations

Track B | Grades 9–12 |

Q1 [Type: MCQ | Difficulty: Easy | QID : 52940]

An AI hallucination occurs when an AI generates information that sounds confident but is actually _____.

- A. translated incorrectly
- B. factually wrong, fabricated, or nonsensical
- C. too short to be useful
- D. written in a casual tone

Correct Answer: B

Explanation: A hallucination is confident-sounding output that is factually incorrect, fabricated, or refers to things that don't exist. Translation errors, brevity, and tone are not what define a hallucination.

Q2 [Type: MCQ | Difficulty: Easy | QID : 52942]

A confidence score tells you how sure the AI sounds, but it does not measure the _____ of the response.

- A. length
- B. speed
- C. accuracy
- D. language

Correct Answer: C

Explanation: Confidence reflects how strongly the output matches patterns from training data - it has no link to whether the information is true in the real world. A pattern can be both very strong and very wrong (for example, the common myth that Sydney is Australia's capital).

Q3 [Type: MCQ | Difficulty: Medium | QID : 52944]

One reduction strategy is to ask the AI for _____ - for example, citations, references, or links - so you can independently verify its claims.

- A. shorter answers
- B. emojis
- C. sources
- D. opinions

Correct Answer: C

Explanation: "Ask for sources" is one of the four reduction strategies: be specific, provide context, ask for sources, and verify the output. Asking for sources doesn't guarantee they are real - but it gives you something concrete to check, which is the only way to catch source hallucinations.

Q4 [Type: MCQ | Difficulty: Easy | QID : 52946]

AI hallucinations happen because the AI is deliberately trying to mislead or deceive the user.

- A. True
- B. False

Correct Answer: B (False)

Explanation: AI has no intentions. It generates text by predicting probable word sequences from training patterns. When those patterns produce something untrue, that is a hallucination - but there is no awareness, no motive, no "trying." Framing AI as deceptive misrepresents how it actually works.

Q5 [Type: MCQ | Difficulty: Medium | QID : 52947]

If you search for a very specific claim that an AI made - for example, a quoted study or a named expert - and find absolutely no online evidence that it exists, that is a warning sign the claim might be a hallucination.

- A. True
- B. False

Correct Answer: A (True)

Explanation: "No results when you search" is one of the six warning signs - The reasoning: if a study, person, or event were real, something about it would almost certainly exist online. A claim that vanishes when you search for it is a strong signal the AI may have invented it.

Q6 [Type: MCQ | Difficulty: Medium | QID : 52948]

A source-grounded AI tool like NotebookLM, when given a source document, can never produce a hallucination.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Source grounding greatly reduces hallucinations but does not eliminate them. The AI can still misinterpret the source, overgeneralize, blend information from different parts in misleading ways, or emphasize the wrong points. "Less likely" is not the same as "impossible."

Q7 [Type: MCQ | Difficulty: Easy | QID : 52950]

An AI confidently states: "The capital of Australia is Sydney." (The actual capital is Canberra.) Which type of hallucination is this?

- A. Factual
- B. Source
- C. Entity
- D. Relationship

Correct Answer: A

Explanation: A Factual hallucination is the simplest type: a basic fact is wrong. The Sydney/Canberra mix-up is the canonical example because the AI has learned a popular myth as a strong pattern - high confidence, false claim.

Q8 [Type: MCQ | Difficulty: Easy | QID : 52952]

An AI response claims: "According to a 2021 study by the Institute of Imaginary Science, coffee causes levitation." The Institute and the study do not actually exist. What type of hallucination is this?

- A. Source
- B. Relationship
- C. Factual
- D. Logical

Correct Answer: A

Explanation: Source hallucinations happen when the AI fabricates citations, studies, papers, or references that do not exist. This is exactly the failure pattern that got the New York lawyer in trouble - fake court cases that sounded entirely plausible.

Q9 [Type: MCQ | Difficulty: Easy | QID : 52954]

An AI confidently describes "the city of Sunvale, the technology capital of Europe." No such city exists anywhere in Europe. Which type of hallucination is this?

- A. Factual
- B. Logical
- C. Entity
- D. Source

Correct Answer: C

Explanation: Entity hallucinations involve inventing people, places, or things that do not exist - like a fake city, a fictional professor, or an imaginary product. The AI assembles a plausible-sounding name from patterns in its training data, but the entity itself isn't real.

Q10 [Type: MCQ | Difficulty: Easy | QID : 52957]

Why does NotebookLM generally have a lower risk of hallucination compared to a standard instance of ChatGPT?

- A. It is source-grounded and primarily answers based on documents the user provides
- B. It uses a special "Truth Token" that forces it to be honest
- C. It is trained on a much larger dataset than ChatGPT
- D. It is programmed to prioritize speed over accuracy

Correct Answer: A

Explanation: By restricting its search space to the documents you provide, NotebookLM avoids drawing freely from general training patterns. It will even refuse to answer questions it cannot ground in your sources - saying "I don't know" is a feature, not a bug.

Q11 [Type: MCQ | Difficulty: Easy | QID : 52959]

An AI states: "All cats are mammals. My pet is a mammal. Therefore, my pet is a cat." Which type of hallucination is this?

- A. Source
- B. Entity
- C. Logical
- D. Factual

Correct Answer: C

Explanation: Logical hallucinations involve faulty reasoning - the conclusion does not follow from the premises. Many animals are mammals (dogs, rabbits, humans), so being a mammal

does not mean being a cat. The AI strung together true statements but reached an invalid conclusion.

Q12 [Type: MCQ | Difficulty: Medium | QID : 52960]

An AI confidently states: "Einstein invented the telephone." Both Einstein and the telephone are real, but the connection between them is wrong - Alexander Graham Bell invented the telephone. Which type of hallucination is this?

- A. Logical
- B. Relationship
- C. Source
- D. Entity

Correct Answer: B

Explanation: Relationship hallucinations are tricky because every piece of the claim is real (Einstein existed, telephones exist), but the connection between them is invented. These are particularly hard to spot - the true components make the false link feel believable, which is also why "mixes real and fake" is on the warning-signs list.

Q13 [Type: MCQ | Difficulty: Medium | QID : 52961]

In 2024, Air Canada's customer-service chatbot gave a passenger incorrect information about the airline's refund policy. When the passenger tried to claim the refund based on the chatbot's advice, the airline refused. The case went to court. What was the key lesson from this case?

- A. AI chatbots cannot legally be used in customer service
- B. The airline was held legally responsible for the chatbot's hallucinated advice
- C. Customers should not read chatbot responses
- D. AI tools should be banned from the airline industry

Correct Answer: B

Explanation: The Air Canada case is one of the famous examples. The court treated the company as responsible for what its AI told customers. The takeaway parallels the lawyer case: when AI hallucinates, the person or organisation acting on the output is the one held accountable.

Q14 [Type: MCQ | Difficulty: Medium | QID : 52962]

You learned about the "attention" mechanism - a spotlight that helps an AI focus on important words. How does attention relate to hallucinations?

- A. Attention acts as a fact-checker that prevents the AI from being wrong

- B. Attention helps the AI focus on important words to stay fluent and on-topic, but it does not guarantee accuracy
- C. Attention is a manual dial users can turn to reduce hallucinations
- D. Attention is the part of the AI that searches the internet for real sources

Correct Answer: B

Explanation: Attention improves fluency and contextual coherence - it's what makes outputs sound polished and on-topic. But attention is about which words the model weights heavily, not about whether the resulting prediction is true. A confidently wrong answer can still have perfectly tuned attention behind it.

Q15 [Type: MCQ | Difficulty: Hard | QID : 52963]

Two AI responses about the same topic are produced. Response 1 has a confidence score of 92%; Response 2 has a confidence score of 60%. What can you conclude about which is more factually accurate?

- A. Response 1 is more accurate because of the higher score
- B. Response 2 is more accurate because lower confidence means the AI is being more careful
- C. Nothing can be concluded from confidence alone
- D. Both are equally accurate because they come from the same AI

Correct Answer: C

Explanation: Confidence and accuracy are independent. A 92% response could match a strong but mistaken pattern (a popular myth). A 60% response could be correct but based on a rarer or less-represented pattern. Without independent verification, the score alone tells you nothing about truth.

Q16 [Type: MCQ | Difficulty: Hard | QID : 52965]

A student is using AI for something high-stakes - for example, medical advice or specific legal information. Which combination of strategies BEST reduces the risk of acting on a hallucination?

- A. Trust whichever answer sounds most confident
- B. Ask the same question many times until the AI gives consistent answers
- C. Be specific in the prompt, provide context, ask the AI for sources, AND verify the output against a trusted source before acting on it
- D. Avoid giving the AI any context so it has to figure everything out on its own

Correct Answer: C

Explanation: There are four reduction strategies - be specific, provide context, ask for sources, verify the output - and they work best together. None of them alone is enough for high-stakes use, but combined they keep a human firmly in the loop. Confidence (A), consistency (B), and removing context (D) all fail to address the actual cause: pattern matching without understanding.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 52966]

Scenario: A researcher uploads a PDF of a scientific paper to both NotebookLM and ChatGPT. She asks both, "What does the paper say about the side effects of Drug X?" ChatGPT lists side effects that sound medical but are not actually in the paper. NotebookLM lists only the side effects that appear in the paper and offers citations. The researcher concludes: "NotebookLM is source-grounded, so it will never hallucinate as long as I provide a source."

Question: Is her conclusion fully correct?

- A. Yes - source-grounded AI only ever draws from provided sources
- B. No - even with a source, NotebookLM can still misinterpret, overgeneralize, or blend information from different parts of the paper in misleading ways
- C. Yes - hallucination is impossible when an answer can be traced to a source
- D. No - NotebookLM has no internet access, so it cannot verify anything at all

Correct Answer: B

Explanation: Source grounding is a strong defence, but it is not perfect. The model can still misread the paper, overgeneralise from one section, or stitch together unrelated details. The discipline is the same as with any AI output: trust but verify. Option D is a common misunderstanding - NotebookLM does have optional web search; the deeper issue is faithful interpretation, not connectivity.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 52969]

Scenario: A medical chatbot in a hospital triage system tells a patient with chest pain, "You are likely experiencing a panic attack. Take deep breaths and rest." The patient is actually having a heart attack. The chatbot's response was confident and based on pattern matching from training data.

Question: Which underlying cause of hallucination is most directly responsible for this dangerous outcome?

- A. The AI lacks real understanding of medicine
- B. The AI had a complete data gap for heart-attack symptoms
- C. The AI's confidence score was too low, which is why it should not have been deployed
- D. The AI was deliberately programmed to minimise emergency-room visits

Correct Answer: A

Explanation: This is "pattern matching gone wrong due to no real understanding." Chest pain shows up in many contexts in training text - panic attacks, indigestion, heart attacks, anxiety - and the model picks the statistically common framing without clinical reasoning. High confidence here is part of the danger, not a safeguard.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 52972]

Scenario: A junior lawyer used an AI assistant to research case precedents and submitted a court filing that included AI-provided citations. Opposing counsel pointed out that several cited cases did not exist - the AI had hallucinated them. The lawyer faced serious professional consequences.

Question: What should the law firm change about how they allow AI to be used for legal research?

(Respond in 3–4 sentences.)

Model Answer: The firm should require that every citation produced by an AI be independently verified in a legal database before being included in any filing - AI should be treated as a starting point, not a final source. They should create explicit policies stating that lawyers remain professionally responsible for verifying AI-assisted research, since the legal consequences of fake citations fall on the lawyer and the firm, not on the AI tool. Training should also cover the specific risk of source hallucinations, so junior staff understand that fabricated cases sound exactly as plausible as real ones. The underlying principle is the same as in the Air Canada case: when AI hallucinates, the human or organisation using it is the one held accountable.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 52973]

Scenario: A high school is considering integrating an AI tutor into the curriculum. The AI can answer student questions, provide explanations, and check work. Some teachers are excited about reducing workload; others worry about hallucinations leading students to learn incorrect information.

Question: Identify ONE significant benefit AND ONE significant risk, and describe what the school should require to balance them.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that an AI tutor can provide individualised attention and immediate explanations at any hour, which a single classroom teacher cannot offer to every student simultaneously. A significant risk is that AI tutors can hallucinate facts, dates, or concepts and present them with the same confidence as correct information - and students, still building their understanding, often lack the background to recognise the errors. To balance these, the school should require AI outputs to be paired with verified textbook material for factual content, have teachers periodically audit the AI's coverage for mistakes, and explicitly teach students to verify AI claims using the warning-signs and reduction strategies rather than

accept them at face value. In short: AI assists, the teacher reviews, and the student stays in the loop.